



Lee HyoIn

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LANGUAGES

English

SKILLS

- Teamwork & Collaboration
- Problem-Solving
- Programming(ROS2, C++, Python)

EDUCATION

Chungnam University

2019 — 2026

Daejeon | BA in mecahtroics

Total GPA of 4.04 / 4.5, Major GPA of 4.06 / 4.5

INTERNSHIPS

undergraduate research student

Jan 2026 — Jul 2026

intelligent Control & Interactive Robotics Lab - Daejeon, S.Korea

- Conducted reviews and seminars on Vision-Language-Action (VLA) models
- Studied robot control algorithms by integrating Pinocchio with ROS2 frameworks.
- Researched control theories, including Impedance and Admittance control, for contact-rich tasks.

Intern

Feb 2024 — Jun 2024

Burnyoung - Daejeon, S.Korea

- Developed a maintenance robot for underground water supply systems
- Designed a sheet-fed dispensing mechanism for robotic material handling
- Designed and modeled a tracked platform using Autodesk Fusion 360

PROJECT EXPERIENCE

Implementation of Imitation Learning via Admittance Control Daejeon, S.Korea

Undergraduate Researcher Mar. 2026 – Jul. 2026

- Applied Admittance Control by estimating external forces based on the current values of motors.
- Calculated real-time gravity compensation torque utilizing the MuJoCo physics engine.
- Significantly reduced communication latency by establishing direct control via the Dynamixel SDK.

Development of a Multi-Floor Delivery Robot utilizing Imitation

Learning Daejeon, S.Korea

Team Leader / Member Jan. 2025 – Nov. 2025

- Implemented Task-Level Autonomous Navigation using the ROS2 Nav2 package.
- Analyzed and controlled dual-arm robot movements using DH-parameters, forward kinematics, and inverse kinematics.
- Applied a lightweight YOLOv5 model for real-time elevator button recognition.
- Executed precise robotic arm control by generating paths with a Cubic Trajectory algorithm.

Control of a 2-DOF Robotic Arm on a Differential Drive Mobile Robot

Daejeon, S.Korea

Team Member Mar. 2025 – Jun. 2025

- Conducted dynamic system simulations utilizing MATLAB Simulink.
- Implemented model-based PID control relying on the precise dynamic equations of the robot.
- Generated smooth motion trajectories utilizing the LSPB (Linear Segment with Parabolic Blends) algorithm.

Image Stitching via Digital Image Processing Daejeon, S.Korea

Individual Project Sep.2024 – Dec.2024

- Implemented the core SIFT (Scale-Invariant Feature Transform) algorithm, encompassing Gaussian pyramid construction, DoG generation, precise keypoint detection, and descriptor extraction.
- Executed robust feature matching across image pairs by utilizing descriptor distance comparisons and ratio test filtering.
- Estimated optimal homography matrices utilizing the RANSAC algorithm to effectively eliminate outliers and accurately align overlapping image regions.
- Generated seamless panoramic images by applying perspective warping and composing transformed image segments onto an integrated canvas.

RecurDyn-based Item Sorting Simulation Daejeon, S.Korea

Team Member Mar.2024 – Jul.2024

- Conducted a dynamic simulation utilizing a UR3 robotic arm to classify and sort items of diverse geometric shapes.
- Generated and controlled precise robotic motion trajectories using the Expression List feature in RecurDyn.
- Optimized contact parameters to ensure stable and realistic physical interactions during the manipulation and sorting process.

ACHIEVEMENTS

- **2025.08 High-Tech Robot Industry Director's Award**, for development of a multi-floor delivery robot utilizing Imitation Learning *Chungnam Nat'l Univ.*
- **2025.08 Department Chair's Award**, for successful demonstration of an Imitation Learning-based robotic sequence *Chungnam Nat'l Univ.*
- **2022.08 Excellence Award (Silver Prize)**, MAZE Line Tracer competition (Step Motor Freshman Division) *Dankook Univ.*

CERTIFICATIONS AND LICENSES

- 2025.11 - OPic IH
- 2024.03 - ADsP
- 2023.12 - SQLD
- 2021.10 - Computer Specialist in Spreadsheet & Database Level-1